

Lecture 1: Two Paradigms — A First Look

Non-asymptotic vs. Asymptotic Statistics

Contents

1	Motivation and the Central Question	1
1.1	The goal of statistics: population vs. sample	1
1.2	A motivating example	2
2	The Asymptotic Paradigm: LLN and CLT	3
2.1	Law of Large Numbers	3
2.2	Central Limit Theorem	4
2.3	Consequences: confidence intervals and the asymptotic paradigm in practice	5
3	The Non-Asymptotic Paradigm: First Tools	5
3.1	Markov's inequality	5
3.2	Chebyshev's inequality	6
3.3	First comparison: Chebyshev vs. CLT	6
3.4	Beyond Chebyshev: a preview of exponential bounds	7
4	First Comparison: The Two Paradigms Side by Side	8
5	Summary	8
6	Exercises	9

1 Motivation and the Central Question

1.1 The goal of statistics: population vs. sample

Before stating the central question of this course, let us recall what statistics is trying to do in the first place.

We are interested in some quantity describing an entire *population* — for instance, the average height of all adults in a country, or the probability p that a manufactured item is defective. Mathematically, we model the population by a probability distribution P , and the quantity of interest is a feature of P :

$$\theta = \theta(P), \quad \text{e.g. } \theta = \mathbb{E}[X] \text{ (the population mean), where } X \sim P.$$

The fundamental difficulty is that P — and hence θ — is *unknown*: we can never examine the whole population. What we *can* do is observing a finite **sample** drawn from it,

$$X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} P,$$

and compute from these observations an **estimator** (a statistic) $\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n)$, e.g. the sample mean $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ as an estimator of $\mu = \mathbb{E}[X]$.

Population vs. Sample

Population (unknown)	Sample (observed)
distribution P	data $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P$
parameter $\theta = \theta(P)$ (fixed)	estimator $\hat{\theta}_n$ (random)
e.g. $\mu = \mathbb{E}[X]$	e.g. $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

This is where the sample size n enters. The estimator $\hat{\theta}_n$ is a random variable: a different sample would give a different value. So the error $\hat{\theta}_n - \theta$ is random, and the only honest questions we can ask about it are probabilistic ones. Intuitively, more data should mean a better estimate, and the entire theory in this course is about making that intuition precise: *how does the (random) error $\hat{\theta}_n - \theta$ behave as a function of n ?*

Statistical theory is largely concerned with one question: *how well does a statistic computed from n observations approximate its population counterpart?* The two dominant paradigms answer this in fundamentally different ways.

Central Question of the Course

Asymptotic paradigm: What happens to $\hat{\theta}_n$ as $n \rightarrow \infty$?

Non-asymptotic paradigm: For a *fixed* n , how large can $|\hat{\theta}_n - \theta|$ be with high probability?

Neither question is better in the abstract. They are complementary lenses, each with strengths and failure modes that we will develop across all four lectures.

1.2 A motivating example

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ and $\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

- **Asymptotic answer:** By the Central Limit Theorem,¹

$$\sqrt{n}(\hat{p}_n - p) \xrightarrow{d} \mathcal{N}(0, p(1-p)),$$

¹Here $\xrightarrow{d}$ denotes *convergence in distribution*, defined formally in Definition 2.3. Informally, the CDF of the left-hand side approaches that of the normal limit as $n \rightarrow \infty$.

so an approximate 95% confidence interval is $\hat{p}_n \pm 1.96\sqrt{\hat{p}_n(1 - \hat{p}_n)/n}$.

- **Non-asymptotic answer:** By the Hoeffding inequality (to be proved in Lecture 3),

$$\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2) \quad \text{for all } n \geq 1, \varepsilon > 0.$$

Key Observation

The CLT confidence interval requires n to be “large enough.” But how large? The Hoeffding bound holds for *every* n with no approximation error, but may be more conservative than the CLT interval when n is indeed large.

2 The Asymptotic Paradigm: LLN and CLT

2.1 Law of Large Numbers

Definition 2.1 — Convergence in Probability

A sequence of random variables $(Z_n)_{n \geq 1}$ converges **in probability** to Z , written $Z_n \xrightarrow{p} Z$, if

$$\forall \varepsilon > 0, \quad \lim_{n \rightarrow \infty} \mathbb{P}(|Z_n - Z| > \varepsilon) = 0.$$

Theorem 2.2 — Weak Law of Large Numbers (WLLN)

Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$. Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then

$$\bar{X}_n \xrightarrow{p} \mu.$$

Proof. By Chebyshev’s inequality (proved in Section 3),

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

□

Remark — Assumptions

The proof uses only $\sigma^2 < \infty$. The WLLN holds under the weaker assumption $\mathbb{E}[|X_1|] < \infty$ (Kolmogorov’s WLLN), which we do not prove here. The Strong LLN ($\bar{X}_n \xrightarrow{\text{a.s.}} \mu$) additionally requires careful measure-theoretic arguments.

2.2 Central Limit Theorem

Definition 2.3 — Convergence in Distribution

(Z_n) converges **in distribution** to Z , written $Z_n \xrightarrow{d} Z$, if

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n \leq t) = \mathbb{P}(Z \leq t)$$

at every continuity point t of the CDF of Z .

Theorem 2.4 — Central Limit Theorem (CLT)

Let $X_1, X_2, \dots \stackrel{\text{iid}}{\sim}$ with $\mathbb{E}[X_1] = \mu$, $\text{Var}(X_1) = \sigma^2 \in (0, \infty)$. Then

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

We sketch the standard proof via moment generating functions (MGFs) under the additional assumption that $M_X(t) = \mathbb{E}[e^{tX}]$ exists in a neighborhood of zero.

Proof sketch via MGF. Center and scale: let $Y_i = (X_i - \mu)/\sigma$, so $\mathbb{E}[Y_i] = 0$, $\mathbb{E}[Y_i^2] = 1$. Let $S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$. We show $M_{S_n}(t) \rightarrow e^{t^2/2}$ for all t .

Step 1 (Independence). Since the Y_i are i.i.d.,

$$M_{S_n}(t) = \mathbb{E} \left[\prod_{i=1}^n e^{\frac{t}{\sqrt{n}} Y_i} \right] = \prod_{i=1}^n \mathbb{E} \left[e^{\frac{t}{\sqrt{n}} Y_i} \right] = \left[M_Y \left(\frac{t}{\sqrt{n}} \right) \right]^n.$$

Step 2 (Taylor expansion of M_Y around 0). Because M_Y exists in a neighborhood of zero, it is infinitely differentiable there, and its derivatives at zero recover the moments: $M_Y^{(k)}(0) = \mathbb{E}[Y^k]$. In particular,

$$M_Y(0) = 1, \quad M_Y'(0) = \mathbb{E}[Y] = 0, \quad M_Y''(0) = \mathbb{E}[Y^2] = 1.$$

By Taylor's theorem with Lagrange remainder, for s in a neighborhood of zero,

$$M_Y(s) = M_Y(0) + M_Y'(0)s + \frac{M_Y''(0)}{2}s^2 + \frac{M_Y'''(\xi)}{6}s^3 = 1 + \frac{s^2}{2} + \frac{M_Y'''(\xi)}{6}s^3$$

for some ξ between 0 and s . Since M_Y''' is continuous, it is bounded on a closed interval around zero, say $|M_Y'''| \leq C$ there, so the remainder term is at most $\frac{C}{6}|s|^3$. Now fix t and substitute $s = t/\sqrt{n}$ (which lies in this neighborhood once n is large enough):

$$M_Y \left(\frac{t}{\sqrt{n}} \right) = 1 + \frac{t^2}{2n} + r_n, \quad |r_n| \leq \frac{C|t|^3}{6n^{3/2}} = O(n^{-3/2}).$$

Step 3 (Taking the limit). Combining Steps 1 and 2 and taking logarithms, using $\log(1+x) = x + O(x^2)$ as $x \rightarrow 0$:

$$\log M_{S_n}(t) = n \log \left(1 + \frac{t^2}{2n} + r_n \right) = n \left(\frac{t^2}{2n} + O(n^{-3/2}) \right) = \frac{t^2}{2} + O(n^{-1/2}) \rightarrow \frac{t^2}{2}.$$

Hence $M_{S_n}(t) \rightarrow e^{t^2/2}$, the MGF of $\mathcal{N}(0, 1)$. Since MGF convergence on an interval implies convergence in distribution, the result follows. $\square$

2.3 Consequences: confidence intervals and the asymptotic paradigm in practice

The CLT underpins most classical statistical procedures.

Example 2.5 — Asymptotic Confidence Interval

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ with mean μ and variance σ^2 , an approximate $(1 - \alpha)$ confidence interval for μ is

$$\bar{X}_n \pm z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the $(1 - \alpha/2)$ quantile of $\mathcal{N}(0, 1)$ and $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$. The coverage $\rightarrow 1 - \alpha$ as $n \rightarrow \infty$; for finite n the actual coverage may differ.

Remark — What Does “Asymptotic” Mean Practically?

“ n large enough” is context-dependent:

- For near-Gaussian data: $n \geq 30$ often suffices.
- For heavy-tailed distributions: hundreds or thousands.
- In high dimensions ($p \approx n$): the CLT may never be accurate.

Quantifying *how large* is precisely the project of non-asymptotic statistics.

3 The Non-Asymptotic Paradigm: First Tools

3.1 Markov’s inequality

Theorem 3.1 — Markov’s Inequality

Let $Z \geq 0$ be a non-negative random variable with $\mathbb{E}[Z] < \infty$. Then for all $t > 0$,

$$\mathbb{P}(Z \geq t) \leq \frac{\mathbb{E}[Z]}{t}.$$

Proof. Write $Z \geq t \cdot \mathbf{1}\{Z \geq t\}$ pointwise. Taking expectations: $\mathbb{E}[Z] \geq t \cdot \mathbb{P}(Z \geq t)$. Divide both sides by $t > 0$. $\square$

Remark — Sharpness

Markov's inequality is tight: for $Z = t$ almost surely, both sides equal 1. In general the bound is loose because it uses only the first moment.

3.2 Chebyshev's inequality**Theorem 3.2 — Chebyshev's Inequality**

Let Z be a random variable with $\mathbb{E}[Z] = \mu$ and $\text{Var}(Z) = \sigma^2 < \infty$. For all $\varepsilon > 0$,

$$\mathbb{P}(|Z - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Apply Markov's inequality to the non-negative random variable $(Z - \mu)^2$ with threshold ε^2 :

$$\mathbb{P}((Z - \mu)^2 \geq \varepsilon^2) \leq \frac{\mathbb{E}[(Z - \mu)^2]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}. \quad \square$$

Corollary 3.1 (Non-asymptotic bound for the sample mean). *Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ with mean μ and $\text{Var}(X_1) = \sigma^2 < \infty$. Then*

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

In particular, setting the right-hand side equal to δ gives $\varepsilon = \sigma/\sqrt{n\delta}$, so with probability at least $1 - \delta$,

$$|\bar{X}_n - \mu| \leq \frac{\sigma}{\sqrt{n\delta}}.$$

3.3 First comparison: Chebyshev vs. CLT**Example 3.3 — Sample Mean, $\sigma = 1$, $\delta = 0.05$**

How many samples n do we need so that $|\bar{X}_n - \mu| \leq 0.1$ with probability ≥ 0.95 ?

Chebyshev (Corollary 3.1): $n \geq \sigma^2/(\varepsilon^2\delta) = 1/(0.01 \times 0.05) = 2000$.

CLT (asymptotic approximation): $z_{0.025} \approx 1.96$, so we need $1.96/\sqrt{n} \leq 0.1$, giving $n \geq 385$.

Interpretation of Example 3.3

- The CLT bound ($n = 385$) is much tighter *because* it exploits the Gaussian approximation, which requires n to be large.
- The Chebyshev bound ($n = 2000$) is valid for *every distribution* with $\sigma^2 < \infty$, with no approximation.
- The Chebyshev bound decays as $O(1/n)$; the CLT tail decays much faster (exponentially in $\varepsilon^2 n$). This gap motivates concentration inequalities (Lectures 2–3).

3.4 Beyond Chebyshev: a preview of exponential bounds

Chebyshev uses the second moment. What if we use *all* moments via the MGF? For $t > 0$, by Markov applied to e^{tZ} :

$$\mathbb{P}(Z \geq \varepsilon) = \mathbb{P}(e^{tZ} \geq e^{t\varepsilon}) \leq \frac{\mathbb{E}[e^{tZ}]}{e^{t\varepsilon}} = e^{-t\varepsilon} M_Z(t).$$

Optimizing over $t > 0$ gives the **Cramér–Chernoff bound**, which yields *exponentially* decaying tails rather than $O(1/n)$. This is the gateway to concentration inequalities — the subject of Lecture 2.

Preview: Hoeffding’s Inequality

For bounded i.i.d. $X_i \in [a_i, b_i]$, the Cramér–Chernoff approach yields

$$\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon) \leq \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

For $X_i \in [0, 1]$: $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}$, yielding $n \geq \log(2/\delta)/(2\varepsilon^2) \approx 185$ in Example 3.3 — much better than Chebyshev.

4 First Comparison: The Two Paradigms Side by Side

	Asymptotic	Non-asymptotic
Central question	What is the limit as $n \rightarrow \infty$?	What can be guaranteed for fixed n ?
Key tool	CLT, δ -method, MLE theory	Concentration inequalities
Output	Approximate CI, p -value, distribution	Explicit probability bound
Assumptions	Often parametric; n large	Often distribution-free; all n
Strength	Sharp, interpretable, model-based	Finite-sample validity, high-dim. friendly
Weakness	Approximation error uncontrolled for finite n	Bounds may be loose (conservative)
Typical decay	CLT CI width $\sim \sigma/\sqrt{n}$	Chebyshev $O(1/n)$; Hoeffding $O(e^{-cn})$

Remark — Are They Competitors?

No. Asymptotic theory gives sharp, interpretable results when n is large and the model is correct. Non-asymptotic theory gives *guaranteed* bounds that are valid for all n and require fewer distributional assumptions. In modern statistics and machine learning, both are used: asymptotic results for inference and interpretability; non-asymptotic results for learning-theoretic guarantees and high-dimensional problems.

5 Summary

1. The **WLLN** states $\bar{X}_n \xrightarrow{p} \mu$; proof uses Chebyshev.
2. The **CLT** states $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$; proof uses MGFs.
3. The CLT enables approximate confidence intervals, but the approximation error is *uncontrolled* for finite n .
4. **Markov** and **Chebyshev** give distribution-free bounds valid for all n , but with polynomial ($O(1/n)$) tail decay.
5. The MGF/Cramér–Chernoff method yields **exponential bounds**, closing the gap. This is the program of Lectures 2–3.

Looking ahead. Lecture 2 dives into the asymptotic paradigm more deeply: the MLE's consistency, asymptotic normality ($\sqrt{n}$ -rate), and the Delta method, together with situations where these results break down.

6 Exercises

Exercise 6.1 (Chebyshev applied). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$, so $\mathbb{E}[X_i] = 1/\lambda$ and $\text{Var}(X_i) = 1/\lambda^2$.

- (a) Use Chebyshev's inequality to find the minimum n such that $\mathbb{P}(|\bar{X}_n - 1/\lambda| \geq \varepsilon) \leq \delta$.
- (b) Compare with the CLT-based requirement.
- (c) For which values of n do you expect the CLT approximation to be accurate?

Exercise 6.2 (Tightness of Markov). Construct a non-negative random variable Z such that Markov's inequality is tight, i.e., $\mathbb{P}(Z \geq t) = \mathbb{E}[Z]/t$ for a specific $t > 0$.

Exercise 6.3 (Polynomial vs. exponential tails). Suppose $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq C_1/(n\varepsilon^2)$ (Chebyshev) and $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 2\exp(-2n\varepsilon^2)$ (Hoeffding, $X_i \in [0, 1]$). Fix $\varepsilon = 0.1$ and $\delta = 0.01$. Find the sample size requirement implied by each bound. Which bound is tighter, and by what factor?

Exercise 6.4 (Convergence notions). Prove or give a counterexample:

- (a) $Z_n \xrightarrow{d} c$ (a constant) $\Rightarrow Z_n \xrightarrow{p} c$.
- (b) $Z_n \xrightarrow{p} Z \Rightarrow Z_n \xrightarrow{d} Z$.
- (c) $Z_n \xrightarrow{d} Z \Rightarrow Z_n \xrightarrow{p} Z$ in general.

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